Microlocal Morse Theory and Truncated Microsupport

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Preface

In these notes, We study microlocal Morse theory, which is developed by Guillermou, Kashiwara, and Schapira in [GKS12], but replace microsupports with truncatd microsupprts introduced by Kashiwara, Monteiro-Fernandes, and Schapira in [KMS03].

1 Truncated Microsupport

Definition 1.1. (i) Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}_k(F)$ of T^*X is defined by: $p \notin \mathrm{SS}_k(F)$ if and only if F satisfies the equivalent conditions in Prop.

(ii) aa

2 Results

Proposition 2.1. Let F in $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi\colon M \to \mathbf{R}$ be a real-valued function of class C^1, proper on $\mathrm{supp}(F)$. let $a < b$ be real numbers. We assume $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t \coloneqq \psi^{-1}(]-\infty, t[)$. Then, restriction morphisms

$$H^j(M_b; F) \to H^j(M_a; F)$$

are isomorphic for each $j \leqq k$.

Proof. First, we have

$$H^j(M_t; F) \cong H^j\,\mathrm{R}\Gamma(]-\infty, t[\,; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$, and by the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1}\,\mathrm{SS}_k(F)$, we have

$$\left(H^j\,\mathrm{R}\Gamma_{[a,b[}\,\mathrm{R}\psi_* F\right)_{\psi(x)} = 0$$

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for each $j \leqq k$. Therefore, letting $d \in \mathbf{Z}$ be the minimum number that

$$H^d \, \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \neq 0,$$

by the long exact sequence

$$\begin{aligned}
0 &\to H^d \, \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \to H^d \, \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \to H^d \, \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \\
&\to H^{d+1} \, \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \to H^{d+1} \, \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \to H^{d+1} \, \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \\
&\to \cdots \\
&\to H^k \, \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \to H^k \, \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \to H^k \, \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \\
&\to H^{k+1} \, \mathrm{R}\Gamma_{[a,b[}(\mathrm{R}\psi_* F) \to \cdots,
\end{aligned}$$

we have isomorphisms

$$\left(H^j \, \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F)\right)_{\psi(x)} \xrightarrow{\sim} \left(H^j \, \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F)\right)_{\psi(x)}$$

for any $j \leqq k$ and $a \leqq \psi(x) < b$, and then,

$$H^j \, \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F) \xrightarrow{\sim} H^j \, \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F) \tag{2.1}$$

for all $j \leqq k$. Then we can conclude that

$$H^j(M_b \, ; F) \xrightarrow{\sim} H^j(M_a \, ; F)$$

by applying to the equality (2.1) the functor $\Gamma(]-\infty, b[\, ; \ \cdot \ )$ and calculating as follows:

$$\begin{aligned}
\Gamma(]-\infty, b[\, ; H^j \, \mathrm{R}\Gamma_{]-\infty,b[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty,b[}(]-\infty, b[\, ; \mathrm{R}^j\psi_* F) \\
&\cong \Gamma(]-\infty, b[\, ; \mathrm{R}^j\psi_* F) \\
&\cong H^j(\psi^{-1}(]-\infty, b[) \, ; F) \\
&\cong H^j(M_b \, ; F)
\end{aligned}$$

and

$$\begin{aligned}
\Gamma(]-\infty, b[\, ; H^j \, \mathrm{R}\Gamma_{]-\infty,a[}(\mathrm{R}\psi_* F)) &\cong \Gamma_{]-\infty,a[}(]-\infty, b[\, ; \mathrm{R}^j\psi_* F) \\
&\cong \Gamma(]-\infty, a[\, ; \mathrm{R}^j\psi_* F) \\
&\cong H^j(\psi^{-1}(]-\infty, a[) \, ; F) \\
&\cong H^j(M_a \, ; F).
\end{aligned}$$

□

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